

"Falcon 2.0T turbocharger oil-feed coking causing boost loss and underboost"

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Area: bulletins

1 Condition

Some Falcon 2.0T vehicles may exhibit a loss of boost, reduced power under acceleration, and rough running that can set misfire code P0300. In affected units the turbocharger assembly, part number TBO-3470, exhibits restricted bearing lubrication leading to shaft drag and underboost. Drivers may report a whine, hesitation, or MIL illumination during sustained highway loads. See the Falcon Engine Service Manual :: Forced Induction (Turbocharger) for normal boost behavior.

2 Cause

The condition is caused by oil-feed coking that restricts the lubrication passage to turbocharger TBO-3470, the result of degraded oil and a blocked or fatigued oil-feed kit, part number OFK-3420. Carbon deposits in the feed line starve the center bearing housing, increasing friction and reducing achievable boost. The Falcon Engine Service Manual :: Lubrication and Crankcase Ventilation describes the oil supply circuit feeding the turbocharger.

3 Diagnostic Procedure

Confirm the boost complaint and check for stored or current P0300 using the Falcon Engine Service Manual :: Misfire Diagnosis , then cross-reference the related Harrier Engine Service Manual :: Misfire Diagnostic Trouble Code Reference for shared code definitions. Inspect turbocharger TBO-3470 for shaft play and oil-feed restriction, and verify actual boost against commanded boost with a scan tool. Follow the Harrier Engine Service Manual :: Misfire Diagnosis Procedure to rule out ignition and fuel contributors before condemning the turbocharger. Removing the oil-feed line and inspecting for coking confirms the diagnosis.

4 Repair Procedure

Replace turbocharger TBO-3470 and the oil-feed kit OFK-3420 together following the Falcon Engine Service Manual :: Turbocharger Replacement (PROC-TURBO). Perform an engine oil and filter change per the Falcon Engine Service Manual :: Oil and Filter Service (PROC-OIL) to clear contamination that caused the coking. Prime the new turbocharger oil feed before startup and verify boost recovery on a road test. Refer to the Harrier Engine Service Manual :: Engine System Overview for shared powertrain torque references.